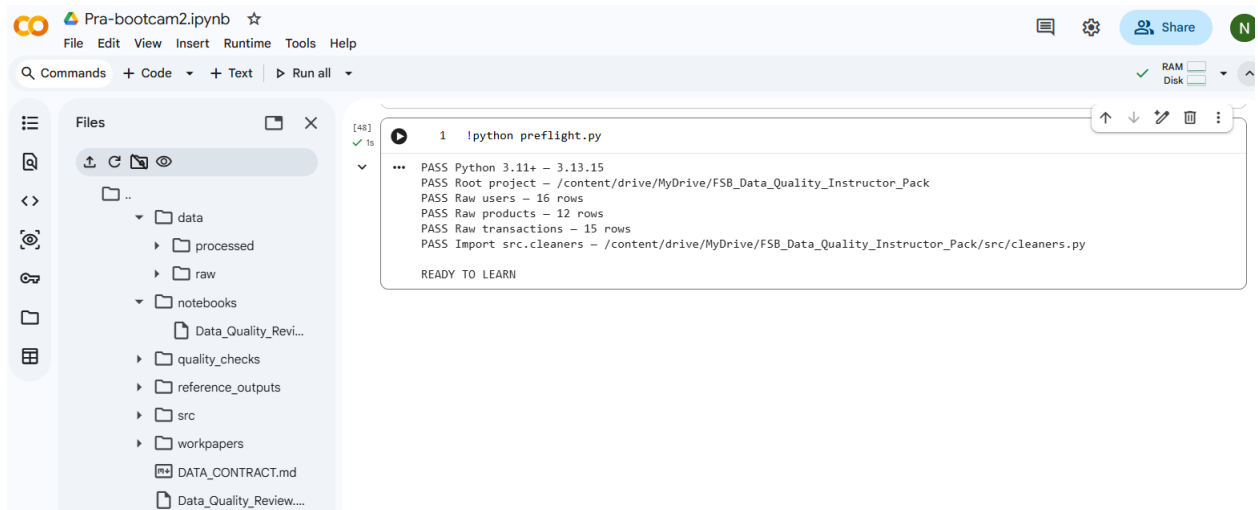


1. Run preflight.py



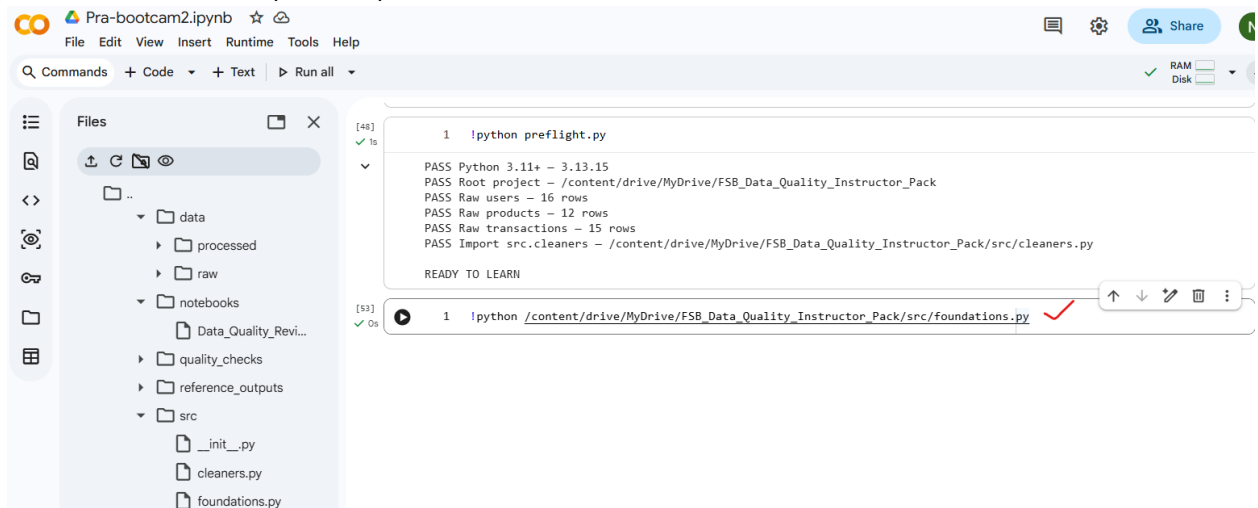
The screenshot shows the JupyterLab interface with the file explorer on the left displaying a directory structure including 'data', 'notebooks', 'quality_checks', 'reference_outputs', 'src', and 'workpapers'. The main code area shows a single cell with the command `!python preflight.py`. The output of the script is displayed below the cell, showing various system checks and file paths.

```
[48] ✓ 1s
1 !python preflight.py

...
PASS Python 3.11+ - 3.13.15
PASS Root project - /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack
PASS Raw users - 16 rows
PASS Raw products - 12 rows
PASS Raw transactions - 15 rows
PASS Import src.cleaners - /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/cleaners.py

READY TO LEARN
```

2. Run – foundations (no error)



The screenshot shows the JupyterLab interface with the file explorer on the left displaying a directory structure including 'data', 'notebooks', 'quality_checks', 'reference_outputs', 'src', and 'workpapers'. The main code area shows two cells. The first cell contains the command `!python preflight.py`. The second cell contains the command `!python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/foundations.py`. The output of the second script is displayed below the cell, showing various system checks and file paths.

```
[48] ✓ 1s
1 !python preflight.py

...
PASS Python 3.11+ - 3.13.15
PASS Root project - /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack
PASS Raw users - 16 rows
PASS Raw products - 12 rows
PASS Raw transactions - 15 rows
PASS Import src.cleaners - /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/cleaners.py

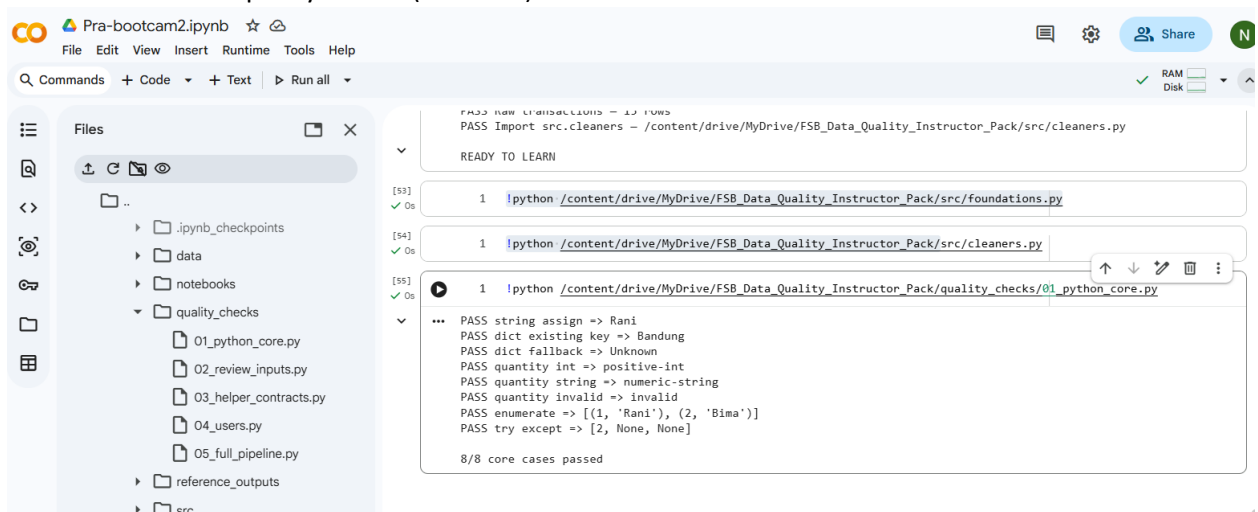
READY TO LEARN

[53] ✓ 0s
1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/foundations.py ✓

...
PASS Python 3.11+ - 3.13.15
PASS Root project - /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack
PASS Raw users - 16 rows
PASS Raw products - 12 rows
PASS Raw transactions - 15 rows
PASS Import src.cleaners - /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/cleaners.py

READY TO LEARN
```

3. Run – cleares & quality checks (no error)



The screenshot shows the JupyterLab interface with the file explorer on the left displaying a directory structure including 'data', 'notebooks', 'quality_checks', 'reference_outputs', 'src', and 'workpapers'. The main code area shows three cells. The first cell contains the command `!python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/foundations.py`. The second cell contains the command `!python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/cleaners.py`. The third cell contains the command `!python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/quality_checks/01_python_core.py`. The output of the third script is displayed below the cell, showing various system checks and file paths.

```
[53] ✓ 0s
1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/foundations.py

[54] ✓ 0s
1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/src/cleaners.py

[55] ✓ 0s
1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/quality_checks/01_python_core.py

...
PASS string assign => Rani
PASS dict existing key => Bandung
PASS dict fallback => Unknown
PASS quantity int => positive-int
PASS quantity string => numeric-string
PASS quantity invalid => invalid
PASS enumerate => [(1, 'Rani'), (2, 'Bima')]
PASS try except => [2, None, None]

8/8 core cases passed
```

4. Run – user.py (no error)

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The file explorer shows a directory structure with files like `01_python_core.py`, `02_review_inputs.py`, `03_helper_contracts.py`, `04_users.py`, and `05_full_pipeline.py`. The code editor shows a list of products, transactions, and users. The output shows the execution of the script, which passes 0/0 core cases.

```
0/0 core cases passed
```

```
[56] 1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/quality_checks/02_review_inputs.py
```

```
... products.json list 12
first {'product_id': 'P001', 'name': ' Laptop Stand ', 'price': '175000', 'category': 'Accessories'}
fields ['category', 'name', 'price', 'product_id']
transactions.json list 15
first {'tx_id': 'T001', 'user_id': 'U001', 'product_id': 'P001', 'quantity': 1, 'tx_date': '2026-08-01'}
fields ['product_id', 'quantity', 'tx_date', 'tx_id', 'user_id']
users.json list 16
first {'user_id': 'U001', 'name': ' Rani Putri ', 'email': 'RANI@MAIL.COM', 'city': 'Bandung'}
fields ['city', 'email', 'name', 'user_id']
```

5. Run – Helper contract (no error)

The screenshot shows a Jupyter Notebook interface with a file explorer on the left and a code editor on the right. The file explorer shows a directory structure with files like `01_python_core.py`, `02_review_inputs.py`, `03_helper_contracts.py`, `04_users.py`, and `05_full_pipeline.py`. The code editor shows a list of products, transactions, and users. The output shows the execution of the script, which passes 11/11 helper cases.

```
11/11 helper cases passed
```

```
[57] 1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/quality_checks/03_helper_contracts.py
```

```
... PASS text None => None
PASS text trim => Rani
PASS text empty => None
PASS email normalize => rani@mail.com
PASS email invalid => None
PASS price string => 19.5
PASS price negative => None
PASS quantity string => 2
PASS quantity zero => None
PASS quantity decimal => None
PASS quantity bool => None
```

6. Run – User.py (no error)

Pra-bootcam2.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Files

- ..
- .ipynb_checkpoints
- data
- notebooks
- quality_checks
 - 01_python_core.py
 - 02_review_inputs.py
 - 03_helper_contracts.py
 - 04_users.py
 - 05_full_pipeline.py
- reference_outputs
- src

```

1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/quality_checks/04_users.py

... PASS count clean
PASS count rejected
PASS conservation
PASS unique IDs
PASS U001 ID trimmed
PASS U001 email normalized
PASS U003 invalid email null
PASS reasons present
PASS decision actions

users: raw=16 clean=14 rejected=2

```

7. Run-Full Pipeline (no error)

Pra-bootcam2.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Files

- ..
- .ipynb_checkpoints
- data
- notebooks
- quality_checks
 - 01_python_core.py
 - 02_review_inputs.py
 - 03_helper_contracts.py
 - 04_users.py
 - 05_full_pipeline.py
- reference_outputs
- src

```

1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/quality_checks/05_full_pipeline.py

... PASS users {'counts': True, 'conservation': True, 'unique IDs': True, 'reasons': True, 'log schema': True}
PASS products {'counts': True, 'conservation': True, 'unique IDs': True, 'reasons': True, 'log schema': True}
PASS transactions {'counts': True, 'conservation': True, 'unique IDs': True, 'reasons': True, 'log schema': True}

```

8.Run – pipeline dan validate_delivery (no error)

Pra-bootcam2.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Files

- ..
- .ipynb_checkpoints
- data
- notebooks
- quality_checks
 - 01_python_core.py
 - 02_review_inputs.py
 - 03_helper_contracts.py
 - 04_users.py
 - 05_full_pipeline.py
- reference_outputs
- src

```

1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/run_pipeline.py

[OK] users: raw=16 clean=14 rejected=2
[OK] products: raw=12 clean=10 rejected=2
[OK] transactions: raw=15 clean=12 rejected=3

1 !python /content/drive/MyDrive/FSB_Data_Quality_Instructor_Pack/validate_delivery.py

... PASS users {'clean count': True, 'rejected count': True, 'conservation': True, 'IDs present': True, 'IDs unique': True}
PASS products {'clean count': True, 'rejected count': True, 'conservation': True, 'IDs present': True, 'IDs unique': True}
PASS transactions {'clean count': True, 'rejected count': True, 'conservation': True, 'IDs present': True, 'IDs unique': True}
PASS user semantics {'U001 exists': True, 'U001 normalized email': True, 'U003 invalid email null': True}
PASS transaction quantities
PASS decision actions/schema {'FIX', 'NULL', 'DROP'}

READY FOR REVIEW

```